FOOD COMPOSITION AND ADDITIVES

Comparison and Assessment of the Difference in Total Dietary Fiber in Cooked Dried Legumes as Determined by Five Methods

ROGER MONGEAU and RENÉ BRASSARD
Health Canada, Food Directorate, Banting Research Centre, Nutrition Research Division, Ottawa, ON, K1A 0L2, Canada

Dried beans (kidney and great northern) and dried peas (chick, green, and yellow) were cooked according to package instructions. Total dietary fiber (TDF) was measured by the Mongeau (AOAC 992.16), Prosky (AOAC 985.29), and Lee (AOAC 991.43) methods (A, B, and C, respectively). Nonstarch polysaccharides (NSP) were measured by the Englyst gas-chromatographic method that included dimethyl sulfoxide treatment, and the lignin measured separately was added to NSP (method D). TDF was also measured by the Li method (E). TDF values ranged from 7 to 48 g/100 g dry weight. Methods B and C gave similar TDF values, which were higher than those from other methods. For kidney beans I, TDF values by methods B and C were up to 2.5 times higher because of the inclusion of starch. The fiber residues from methods A and C contained the same amounts of arabinose, xylose, mannose, galactose, and uronic acid, but glucose was 4 times higher in residue C than in residue A. When α-amylase from porcine pancreas was incorporated in methods B or C for 5 samples, the discrepancies among methods A, B, and C were reduced by 60–98%.

The Mongeau (AOAC **992.16**), Prosky (AOAC **985.29**), and Lee (AOAC **991.43**) methods, respectively identified as A, B, and C, are gravimetric methods that yield similar total dietary fiber (TDF) values for various foods. Good agreement between methods A and B was reported for the determination of TDF in canned legumes (1). However, large amounts of starch were reported in the residues of cooked dried legumes analyzed by methods B and C (2–4). Similarly, TDF values for dried peas differed by a factor of up to 4 among methods because of inconsistencies in the capacity to digest starch (Mongeau and Brassard, unpublished data). Thus, the determination of dietary fiber is critical for cooked dried legumes. Because of the possibility of significant discrepancies for these foods, this methodological problem must be investigated to achieve agreement among dietary fiber methods and to avoid the accumulation of conflicting data.

Received August 24, 1993. Accepted by JL November 10, 1993.

Methods B and C are similar, and it is not known if their starch-digesting capacities can be improved by using additional digestive enzymes rather than dimethyl sulfoxide (DMSO) or heat treatments. DMSO is known to remove starch, but it is not certain if it removes only glucose from the constituent sugars of the residue (5). A comparison of methods A, B, and C with the Englyst gas-chromatographic method plus lignin (method D) permits monitoring of dietary fiber composition. The Li and Cardozo method (3) is a modification of method B: some enzyme treatments are deleted and a larger protein correction is made to the residue (method E). The Li method provides a base to evaluate the proteolytic activity of enzymes.

The aim of the present work was to compare the TDF values for cooked dried legumes determined by the 5 methods and to see if the variability among methods could be improved by using mammalian digestive enzymes.

Experimental

Materials

Dried red kidney beans I (Goya) and II (Townhouse), great northern beans, and chick peas, bought in the Washington, DC, area, were provided by Li and Cardozo (3). The dried green and yellow peas were bought in Ottawa, Canada.

Sample Preparation

Kidney beans I and II, great northern beans, and chick peas were prepared by U.S. Department of Agriculture (USDA) laboratory as described by Li and Cardozo (3). They were quickly soaked (200 g/L deionized water) and cooked for 1 h, according to package instructions. Kidney beans II were cooked for 2 h. They were then freeze-dried, and a portion of each was sent to the Food Directorate laboratory in Ottawa.

The dried green and yellow peas were brought to boiling and simmered for 3 h according to package instructions. For green peas, the package contents were mixed to ensure homogeneity between 2 cooking batches. Exactly the same time and temperature settings were used for the 2 batches. The green peas listed in Tables 1 and 2 were from the first batch, and those listed in Table 3 were from the second batch. For yellow peas, a single large batch was cooked.

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Table 1. TDF in boiled dried green peas: effect of incorporating an APP treatment in methods B and C

Method[a]	TDF,[b] % (dry wt. basis)	Starch in TDF[c]
A with APP at 55°C	9.9 ± 0.1^{w}	2.7
A with APP at 37°C	9.0 ± 0.1^{v}	1.8
B	19.0 ± 0.2^{z}	11.8
B^{+APP}	12.3 ± 0.3^{y}	5.1
C	18.8 ± 0.3^{z}	11.6
C^{+APP}	12.3 ± 0.3^{xy}	5.1
D	7.2 ± 0.2^{u}	0.0
D without DMSO	10.4 ± 0.1^{wx}	3.2

[a] A, rapid method, AOAC **992.16**; B, AOAC **985.29**; C, Tris-Mes method, AOAC **991.43**; C^{APP}, α-amylase from porcine pancreas, for 30 min after heat-stable amylase; D, nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added; E, Li and Cardozo (3). A lignin value of 0.4 was added.
[b] Values are means ± standard deviations for 3 or 4 determinations. Values in this column not sharing the same superscript are significantly ($P < 0.001$) different.
[c] TDF minus TDF by method D.

Dietary Fiber Methods

Total dietary fiber (TDF) was measured by the following methods: A, Mongeau (AOAC **992.16**) (1); B, Prosky (AOAC **985.29**) (6); C, Lee (AOAC **991.43**) (7); D, Englyst GC (8) plus lignin; and E, Li and Cardozo (3). Method E is a simplified Prosky method that includes autoclaving at 130°C and excludes some enzyme treatments. The enzymes used were obtained from the suppliers specified in each method. No substitutes were used.

When α-amylase from porcine pancreas (APP) (Sigma Chemicals Co., Cat. No. A3176) was used with methods B or C, the enzyme was added after treatment of sample with heat-stable amylase. The APP treatment consisted of adding a 5.0% solution of APP in phosphate buffer (0.1M at pH 7.0) for 30 min at 55°C. A volume of 5 mL of the solution was the best compromise between efficiency and economy for methods B and C. Method A includes 2 APP treatments.

Table 2. Effect of APP and protease on amounts of residual protein of methods B and C for boiled dried green peas

Method modifications[a]	Protein,[b] % dry sample
B	5.0 ± 0.0^{v}
B^{+APP}	3.3 ± 0.2^{u}
B^{+APP} protease deleted	5.5
C	6.9 ± 0.1^{w}
C^{+APP}	2.9 ± 0.3^{uv}
C^{+APP} protease deleted	4.9
E without protease	18.7
$E^{120°C}$ without protease	20.0 ± 0.1^{x}

[a] A, rapid method, AOAC **992.16**; B, AOAC **985.29**; C, Tris-Mes method, AOAC **991.43**; C^{APP}, α-amylase from porcine pancreas, for 30 min after heat-stable amylase; D, nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added; E, Li and Cardozo (3). $E^{120°C}$ is the method of Li and Andrews (16).
[b] Values are means ± standard deviations for 2–4 determinations; $n = 1$ when no standard deviation is shown. Values in this column not sharing the same superscript are significantly ($P < 0.001$) different.

Table 3. Constituent sugars of residues of fiber methods C, C^{+APP}, and D for boiled dried green peas

	Amount of sugar, % (dry wt. basis)							
Method[a]	RHA[b]	FUCO	ARA	XYL	MAN	GAL	GLU	URON
C	0.13	0.10	2.66	0.32	0.16	0.62	20.41	1.46
	0.13	0.10	2.46	0.27	0.13	0.56	21.86	1.18
C^{+APP}	0.13	0.09	2.50	0.29	0.16	0.60	6.87	1.35
	0.12	0.09	2.36	0.26	0.16	0.55	7.16	1.25
D[c]	0.09	0.08	2.59	0.28	0.09	0.74	1.75	1.14
	0.12	0.09	2.62	0.28	0.09	0.72	1.55	1.08

[a] Method: nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added
[b] RHA, rhamnose; FUCO, fucose; ARA, arabinose; XYL, xylose; MAN, mannose; GAL, galactose; GLU, glucose; URON, uronic acid.
[c] Green peas used with method D were from the first cooking batch, whereas those used with methods C and C^{+APP} were from the second cooking batch.

Determination of Neutral Sugars and Uronic Acids

The polysaccharide components of fiber residues were analyzed according to the Englyst method (8). After hydrolysis by acid treatments, neutral sugars were measured by gas–liquid chromatography as alditol acetates and uronic acid was measured colorimetrically.

Determination of Lignin

Lignin was measured by 2 methods. Permanganate lignin was measured according to the Robertson and Van Soest procedure as described by Mongeau and Brassard (9): the sample was treated with a neutral detergent, incubated with APP for 65 min, treated with an acid detergent, and then treated mildly with permanganate. These sequential treatments removed most of the nonfiber materials. Permanganate lignin values were added to Englyst nonstarch polysaccharides (NSP) to obtain TDF values.

Klason lignin was the material left after all the enzyme and chemical treatments of the Englyst DMSO method.

Statistical Method

The significance of the effect of APP on TDF analysis of dried green and yellow peas was evaluated by 2-tail unpaired t-test (StatView II, Abacus Concepts, Inc., Berkeley, CA).

Results

Table 4 shows TDF values for beans and chick peas measured by 5 methods. The close agreement of most duplicates indicates good repeatability. The number of analyses was re-

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stricted by the limited amounts of samples available from the USDA laboratory. TDF values ranged from 11% for chick peas to 48% (g/100 g, dry weight) for kidney beans I. The unmodified Prosky (B) and Lee (C) methods yielded similar values, which were 1.5 to 2.3 times higher than those from the Mongeau rapid method (A), the Englyst GC method plus lignin (D) or the Li and Cardozo method (E). Analysis by methods A, D, and E of the kidney and great northern beans gave similar dietary fiber contents (18–22%), which were higher than those of chick peas (11–13%). By contrast, with methods B and C, TDF values for kidney beans I (48%) were higher than those for other samples, including chick peas (24–32%). When APP treatment was incorporated in method C (C^{+AAP}), the TDF value decreased to 32% but remained at least 50% higher than those from methods A and D. For kidney beans II and great northern beans, a closer agreement was seen among methods A, C^{+APP}, D, and E. Not enough chick pea sample remained to test method C^{+APP}. Methods A and E gave slightly higher values than method D (Table 4).

Permanganate and Klason lignin contents represented 10% of nonstarch polysaccharides (NSP) or less. They were respectively 1.77 and 1.65% for kidney beans I, 1.41 and 1.36% for kidney beans II, 0.76 and 0.42% for great northern beans, and 0.88 and 0.56% for chick peas. Klason lignin (y) was slightly lower than permanganate lignin (x), but values were highly correlated ($y = 1.27x - 0.53$, $r = 0.993$). The permanganate lignin value was added to the Englyst NSP to obtain the TDF values by method D in Table 4.

Table 5 compares the constituent sugars of residues from methods A, C, and D for kidney beans I. The compositions were similar for uronic acid and neutral sugars, including arabinose/xylose ratios (2.7–2.9). An exception were values of residual glucose, which were 4.4, 7.6, and 30% for methods D, A, and C, respectively. Method D included a DMSO treatment that dissolved starch, and the 4.4% glucose obtained was part of NSP. This result implies that the residual starch contents for methods A and C were 3.2 and 25.6%, respectively. These figures correspond to the difference between their respective TDFs and the TDF by method D (2 and 29%; Table 4).

For kidney beans II, great northern beans, and chick peas, values for total sugars agreed within 0.2%, but glucose values were higher (7.8–11.3%) by method C than by method D (Table 6). This difference accounted for 70–88% of the difference in the TDF values (11.3–14.1%; Table 4).

Because of the lack of availability of the four samples listed in Tables 4–6, green and yellow peas were used to complement the method comparison for boiled legumes. The TDF contents of boiled green peas ranged from 7.2 to 19% depending on method (Table 1). The starch contents of the respective fiber residues, calculated as the difference in TDF from method D, varied from 2.7% (method A) to 11.8% (method B). The starch content was 1.8% when APP was used at 37°C for 18 h in method A and 3.2% when method D was used without DMSO. When methods B and C were used with an APP treatment, most of the residual starch was digested ($P < 0.001$) and 5.1% remained in the fiber residues (Table 1).

Table 4. Total dietary fiber values for beans and chick peas measured by different methods[a]

	Total dietary fiber, % (dry wt. basis)					
Sample[b]	A	B	C	C^{+APP}	D	E
Kidney beans I	20.7	47.1	47.9	31.1	18.9	
	21.3	44.5	47.3	32.1	18.8	
Kidney beans II	21.3	29.0	31.7	22.1	17.7	21.2
	21.9	30.4	31.5	21.4	17.3	21.9
Great northern beans	20.7	29.3	30.6	23.0	19.4	22.3
	20.7	28.9	31.1	23.4	19.6	22.4
Chick peas	12.7	24.9	23.8		10.7	13.2
	12.8	24.3	23.5		10.7	12.8

[a] A, rapid method, AOAC **992.16**; B, AOAC **985.29**; C, Tris-Mes method, AOAC **991.43**; C^{+APP}, α-amylase from porcine pancreas, for 30 min after heat-stable amylase; D, nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added; E, Li and Cardozo (3).

[b] Samples were soaked and boiled for 1 h except for kidney beans II, which were boiled for 2 h.

Because starch digestion increased after incorporation of an APP treatment between the heat-stable amylase and amyloglucosidase treatments, the relative importance of these amylolytic enzymes was tested. When the heat-stable amylase or the amyloglucosidase treatment was deleted in methods B or C (data not shown), the amounts of starch in the residue exceeded the 5.1% given in Table 1 obtained by methods (B^{+APP} and C^{+APP}) using the 3 amylolytic enzymes. Pancreatin was tested in place of APP, but with the concentrations used, less starch was digested because of the lower amylase activity of the enzyme mixture (data not shown).

In contrast to methods B and C, method E does not include a protease treatment. Consequently, the amounts of protein (N × 6.3) to be subtracted from the residues were higher for method E (>18%) than for methods B and C (<7%) (Table 2). The protein content was expressed in percent sample because the amylolytic activity of APP also decreased residue weights. APP and protease treatments together (method B^{+APP} or C^{+APP}) further decreased ($P < 0.001$) residual protein from 5.0–6.9% to about 3% because of the additional proteolytic activity of APP (Table 2). Removal of the protease treatment from method B^{+APP} or C^{+APP} revealed that the proteolytic activity of APP was equivalent to that of the protease alone. Deletion of protease had no effect on starch contents and TDF values.

A second batch of green peas was cooked for analysis of constituent sugars in residues of methods C and C^{+APP} (Table 3). Although cooking conditions were the same as for the first batch (Tables 1 and 2), the total sugar content reached 26.3% (method C, Table 3) compared with 18.8% TDF (method C, Table 1). This result indicated the poor repeatability of method C for cooked green beans. When incorporated in method C, APP did not change the nonglucose sugars but removed 14.1 g/100 g glucose (starch) and consequently decreased the total sugar content of the residue to 12.0% (Ta-

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Table 5. Constituent sugars of fiber residues of methods A, C, and D for boiled dried kidney beans I[a]

	Amount of sugar, % (dry wt. basis)							
Method	RHA[b]	FUCO	ARA	XYL	MAN	GAL	GLU	URON
A	0.24	0.37	5.55	2.06	0.31	1.37	7.71	3.00
	0.24	0.39	5.63	2.09	0.32	1.40	7.57	3.07
C	0.22	0.33	4.99	1.63	0.49	1.20	31.6	3.23
	0.26	0.34	5.18	1.98	0.48	1.23	28.7	3.77
D	0.21	0.31	5.39	1.86	0.53	1.27	4.35	3.17
	0.20	0.34	5.28	1.82	0.53	1.29	4.38	3.10

[a] Method: nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added.
[b] Abbreviations: RHA, rhamnose; FUCO, fucose; ARA, arabinose; XYL, xylose; MAN, mannose; GAL, galactose; GLU, glucose; URON, uronic acid.

ble 3). The latter value is similar to the TDF value of 12.3%, which includes about 0.4% lignin (Table 1) and indicates that method C^{+APP} is more independent from variations between cooking batches.

A large batch of yellow peas was cooked to permit various analyses on identical material (Tables 7 and 8). TDF values by methods B and C were at least 40% higher than by methods A and D (Table 7). When the APP treatment was performed at 37°C instead of 55°C in method A, TDF did not decrease significantly (–0.8%, $P > 0.01$). When APP was incorporated in methods B and C, the TDF values were 25% lower, and residual starch decreased from 9.8% or more to 3.8% or less. TDF values by methods A, B^{+APP}, and C^{+APP} were within the 16.7–18.0% range. Deleting the DMSO treatment in method D increased the TDF value to 18.5% (Table 7).

The total amount of polysaccharide components in the hydrolyzed residue of method C reached 25.4%, the majority being glucose (17.4%, Table 8). The APP treatment reduced the glucose content to 9.6%. This accounted for >95% of the reduction in total sugars. The remaining glucose content (9.6%) exceeded the nonstarch glucose content in the residue of method D by 4.1% (Table 8), which is close to the 3.8% obtained gravimetrically (Table 7).

Discussion

Dietary fiber is made of complex and heterogeneous polymeric materials that are not easy to separate from other food components, particularly starch (10–16). Methods for dietary fiber have been improved over the past decade, and a good agreement for TDF analyzed by methods A, B, and D was reported for various foods including canned legumes (1). However, it was reported that a substantial amount of starch remains in the insoluble fiber residue of methods B or C for cooked dried legumes (2–4). The present results confirmed the incomplete removal of starch by methods B or C for 6 legumes. Chick peas were not tested with APP but the difference in the glucose contents of the residues of methods C and D (Table 6) accounted for 88% of the gap in the respective TDF values (Table 4). For kidney beans I, APP reduced the discrepancy between methods A and C from 26.6 to 10.6% (Table 4). For kidney beans II, great northern beans, green peas, and yellow peas, APP reduced the mean discrepancy among methods A, B, and C from 9.9 to 1.8% (Tables 1, 3, 4 and 7–8). Overall for the 5 samples tested, APP reduced the discrepancies among methods A, B, and C by 60–98%.

The decreases in the TDF values observed when using APP in methods B or C were attributable only to removal of starch from their residues. This effect could have been demonstrated by treating the residues of methods B and C with starch-digesting enzymes and by measuring the glucose released in the supernatant. However, the release of glucose depends on the enzymatic conditions of the assay. If the enzymatic conditions do

Table 6. Constituent sugars of fiber residues of methods C and D for 3 legumes

		Amount of sugar, % (dry wt. basis)							
Sample	Method[a]	RHA[b]	FUCO	ARA	XYL	MAN	GAL	GLU	URON
Kidney beans II	C	0.20	0.28	4.54	1.76	0.34	1.07	15.4	2.83
		0.23	0.27	4.50	1.78	0.34	1.11	14.6	3.08
	D	0.20	0.28	4.73	1.73	0.57	1.14	4.88	2.73
		0.21	0.27	4.51	1.70	0.54	1.10	4.81	2.71
Great northern beans	C	0.26	0.33	5.68	2.36	0.36	1.05	12.9	3.81
		0.23	0.30	5.35	2.38	0.34	1.00	12.4	3.93
	D	0.23	0.34	5.87	2.21	0.53	1.05	4.81	3.53
		0.20	0.35	5.78	2.24	0.54	1.06	5.03	3.55
Chick peas	C	0.21	0.10	3.44	0.48	0.27	0.58	13.5	2.28
		0.19	0.10	3.45	0.43	0.25	0.54	14.3	1.95
	D	0.20	0.11	3.83	0.46	0.17	0.56	2.56	1.91
		0.21	0.09	3.92	0.45	0.17	0.58	2.54	1.87

[a] Residues of method C were treated as those of method D (nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added).
[b] Abbreviations: RHA, rhamnose; FUCO, fucose; ARA, arabinose; XYL, xylose; MAN, mannose; GAL, galactose; GLU, glucose; URON, uronic acid.

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Table 7. TDF in boiled yellow peas determined by unmodified or modified methods A, B, C, and D

Method[a]	TDF,[b] % (dry wt. basis)	Starch in TDF[c]
A with APP at 55°C	16.7 ± 0.1^{vw}	2.5
A with APP at 37°C	15.9 ± 0.2^{uv}	1.7
B	24.0 ± 0.3^{y}	9.8
B^{+APP}	17.8 ± 0.0^{x}	3.6
C	24.1 ± 0.6^{y}	9.9
C^{+APP}	18.0 ± 0.1^{wx}	3.8
D	14.2 ± 0.1^{u}	0.0
D without DMSO	18.5 ± 1.4^{uvwxy}	4.3

[a] A, rapid method, AOAC **992.16**; B, AOAC **985.29**; C, Tris-Mes method, AOAC **991.43**; C^{APP}, α-amylase from porcine pancreas, for 30 min after heat-stable amylase; D, nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added; E, Li and Cardozo (3). A lignin value of 1.2 was added.

[b] Values are means ± standard deviations for 3 determinations, except for methods B^{+APP} and C^{+APP} ($n = 2$). Values in this column not sharing the same superscript are significantly ($P \leq 0.01$) different.

[c] TDF minus TDF by method D.

not remove all starch, the amount of glucose released in the supernatant will lead to an underestimation of residual starch. On the other hand, if fiber-degrading enzymes are present in the assay, they may release nonstarch glucose, which will lead to overestimation.

In the present work, a different approach was used to estimate the starch in the residue of method C. The residue was hydrolyzed in acidic conditions according to the Englyst GC method. These conditions favor complete depolymerization of fiber polysaccharides into their constituent sugars and minimize sugar degradation. Measuring the sugars by gas chromatography as alditol acetates enables a comparison with NSP constituents determined by the integral Englyst method (8). The enzymes used in the latter method are devoid of fiber-degrading activities, and the DMSO treatment ensures that no starch remains in the residue before acid hydrolysis. This nonstarch glucose value can be subtracted from that found in the residue of the fiber method tested for residual starch. The values for the other constituent sugars of NSP may also permit detection of fiber-degrading enzymes.

Table 8. Constituent sugars of residues of fiber methods C, C^{+APP}, and D for boiled dried yellow peas

Method[a]	Amount of sugar, % (dry wt. basis)							
	RHA[b]	FUCO	ARA	XYL	MAN	GAL	GLU	URON
C	0.23	0.13	3.01	1.29	0.20	0.65	17.35	2.48
	0.22	0.13	2.83	1.39	0.20	0.62	17.45	2.56
C^{+APP}	0.20	0.10	2.99	1.05	0.19	0.63	9.52	2.60
	0.19	0.10	3.00	1.06	0.19	0.63	9.68	2.28
D	0.19	0.10	2.97	1.01	0.12	0.75	5.51	2.22
	0.18	0.11	3.09	1.09	0.11	0.72	5.52	2.25

[a] Residues were treated as those of method D (nonstarch polysaccharides (NSP) according to Englyst GC method using DMSO, with $KMnO_4$ lignin values added).

[b] RHA, rhamnose; FUCO, fucose; ARA, arabinose; XYL, xylose; MAN, mannose; GAL, galactose; GLU, glucose; URON, uronic acid.

APP did not degrade the fiber. For green and yellow peas, APP in method C did not decrease the nonglucose fiber constituents and it did not decrease glucose under the level observed by method D (Tables 3 and 8). For other foods (green beans, carrot, barley, oat bran, high-fiber cereal, and soybran), TDF values obtained gravimetrically by method C with or without APP were similar:

$$C^{+APP} = 0.156 + 0.995(C),\ r^2 = 0.999 \text{ (unpublished data).}$$

APP was incubated at 55°C when used in methods A, B, and C. At 37° or 55°C, the types of enzyme activities of APP are the same but the rate of starch hydrolysis is higher at 55°C. Tables 1 and 7 indicate a slightly better starch digestion at 37°C than at 55°C for method A because a longer incubation was used at 37°C. APP is devoid of fiber-degrading enzyme activities at 37° or 55°C.

Lee and Prosky (5) suggested that DMSO could solubilize hemicelluloses, on the basis of published data including those of Honda et al. (2). However, the present results do not support this hypothesis because the fiber constituent sugars, except glucose, were similar for methods excluding (A and C) and including DMSO (D). There were no definite trends for the 6 legume samples examined (Tables 3, 5, 6, and 8).

APP is an unpurified α-amylase preparation, and it provides a proteolytic activity comparable with that of the protease used in methods B and C (Table 2). This indicates that method B would not take longer if APP replaces the protease, and the capacity to digest starch would be improved. Methods B and C are similar, but the latter uses a Tris-Mes buffer with a temperature-dependent pH, and only one pH adjustment is needed. Because APP works very efficiently at 55°C, the resulting pH would not be compatible with its activity. Thus, if APP replaces the protease in method C, the number of pH adjustments would be the same as in method B.

Unmodified method C included more starch in the fiber residue of the second cooking batch (19.5%, calculated from Table 3) than in the first cooking batch (11.6%, Table 1). However, this inconsistency did not appear when APP was incorporated in method C. The amounts of residual starch were decreased to around 5% for both batches, and the dietary fiber values excluding lignin were close (11.9%, Table 1; 12.0%, Table 3).

Methods A, C, and D showed remarkable agreement in the polysaccharide constituents of dietary fiber, glucose excepted (Tables 3, 5, 6, and 8). Therefore, solving the problem of starch digestion should decrease most of the discrepancy among methods.

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FOOD COMPOSITION AND ADDITIVES

Evaluation of a Gas Chromatographic Method to Identify Irradiated Chicken, Pork, and Beef by Detection of Volatile Hydrocarbons

GEORG A. SCHREIBER, GRIT SCHULZKI, ALMUTH SPIEGELBERG, NORBERT HELLE, and KLAUS W. BÖGL
Federal Institute for Health Protection of Consumers and Veterinary Medicine, General-Pape-Str. 62, D-12101 Berlin, Germany

A method suitable for routine application was used in an interlaboratory study to detect irradiation treatment of chicken carcass, pork, and beef. By using gas chromatographic analysis, 17 participating laboratories determined the quantity of 4 different radiation-induced volatile hydrocarbons (tetradecene, pentadecane, hexadecadiene, and heptadecene) in the fat fraction of coded specimens approximately 3 and 6 months after irradiation. The specimens of each type of meat were supplied by 2 different producers. The dose range tested (0.6–7.5 kGy) included levels commercially used to reduce the number of contaminating microorganisms (1–5 kGy). The method employed permitted a correct identification of irradiated or nonirradiated in 98.3% of the 864 specimens.

Received May 12, 1993. Accepted by JL September 27, 1993.

Treatment of certain types of food by ionizing radiation to increase the shelf life and to reduce health hazards that might be caused by contaminating microorganisms is permissible in a number of countries; in other countries, ionizing radiation is banned. To meet the different national regulations in international trade and to ensure the consumer's free choice to buy irradiated or nonirradiated products, labelling control has become necessary. Various interlaboratory studies have already been performed to evaluate methods that identify irradiation treatment of foods (1–6). The present interlaboratory study describes the gas chromatographic (GC) detection of volatile hydrocarbons (HC) in the fat fraction of raw chicken carcass, pork, and beef. The method was used to identify an irradiation of these products in the standard commercial dose range (1–5 kGy). The method was used for a qualitative judgment only, not for dose estimations.

At the end of the 1960s, details about the detection of irradiated foodstuffs on the basis of changes in the fat fraction were published (7–11). In recent years, interest in the identification of irradiated fatty foodstuffs has grown steadily (12–29). The